Revision

We thank the Recommender and both reviewers for their careful reading of our manuscript and their constructive feedback, which has substantially improved its transparency, in line with the aims of the Registered Report format.

In sum, the main analysis is now the fully registered analysis (N = 295). The deviations from the registered protocol are reported separately, and their consequences are evaluated in sensitivity analyses. We have also applied the registered rule for collapsing conditions, corrected the error in Figure 2, and reported all registered tests in the manuscript (Table 3). The conclusions for the primary outcome (error rate) were the same in all versions of the analysis. For the secondary outcome (IES), the evidence was mixed and the classification of some tests varied between versions. We now state all conclusions more cautiously, distinguishing evidence for the null from inconclusive results and restricting them to the illusions for which evidence for the null was obtained.

Changes to the manuscript are highlighted in bold in the revised manuscript and quoted below where relevant.

Managing board

Your article, entitled **Testing the Relationship between Phenomenological Control related to Illusion Sensitivity**, has now been reviewed. The referees' comments and the recommender's decision are shown below. As you can see, the recommender suggests revisions.
We shall, in principle, be happy to recommend your article as soon as it has been revised in response to the points raised by the referees. When revising your article, we remind you that:
1) Data must be available to readers, either in the text or through an open data repository such as Zenodo (free), Dryad (paid) or an institutional repository. Data must be reusable, thus metadata or accompanying text must carefully describe the data;

All data are openly available in the project's GitHub repository (https://github.com/RealityBending/IllusionGamePhenomenologicalControl), which is cited in the manuscript (Data Availability section). The GitHub repository's README describes each data file (raw and cleaned), for the fully registered analysis and the unregistered sensitivity analysis.

To ensure that a permanent copy remains available to readers, the same files and materials are also archived on Zenodo (DOI: https://doi.org/10.5281/zenodo.23943901) and this archive is cited in the Data Availability section of the manuscript.

2) Details on quantitative analyses (e.g., data treatment and statistical scripts in R, bioinformatic pipeline scripts, etc.) and details concerning simulations (scripts, code) must be available to readers in the text, as appendices, or through an open data repository, such as Zenodo, Dryad or an institutional repository. The scripts or code must be carefully described so that they can be reused;

All data cleaning and analysis scripts (R and Quarto) are available in the same repository and in its Zenodo archive. The shared data files contain the anonymised participant- and trial-level data extracted from the raw experiment files, which cannot be shared as they contain participant identifiers.

The GitHub README describes the order in which the scripts should be run and what each one does (data cleaning and analysis), and lists the software and R packages needed to reproduce the analyses,

with the commands to do so. It also includes a table labelling every analysis step as registered, unregistered (sensitivity analysis) or exploratory, with the section of the script in which it is implemented.

The rendered Data Cleaning and Data Analysis reports, linked from the README, document and label each data cleaning and analysis step.

3) Details on experimental procedures must be available to readers in the text or as appendices;

The experimental procedure is described in full in the Methods section (Participants, Procedure, Phenomenological Control Scale and Illusion Game), including the order and counterbalancing of the tasks, the number and structure of trials, the manipulation of illusion strength and task difficulty, the instructions, and the attention checks.

In addition, all study materials are openly available in the GitHub repository and in its Zenodo archive: the full experiment code, which can also be run in a browser, and the stimuli; the PCS script, scoring guide, audio recordings and rating scales (documents/PCS_materials/; the online implementation of the PCS is available at https://github.com/RealityBending/JSmisc); and the information sheet, consent form, instructions, questionnaires and debriefing used in the study, including screenshots of each screen (documents/ethics/).

4) Authors must have no financial conflict of interest relating to the article. The article must contain a "Conflict of interest disclosure" paragraph before the reference section containing this sentence: "The authors of this article declare that they have no financial conflict of interest with the content of this article."; This disclosure has to be completed by a sentence indicating, if appropriate, that some of the authors are PCI recommenders: “X is a recommender at PCI RR.”

We have added a “Conflict of interest disclosure” section before the references, containing the required sentence (it is also included in the author note). None of the authors is a PCI RR Recommender.

Lines 606 - 607:
“The authors of this article declare that they have no financial conflict of interest with the content of this article.”

Recommender

I have now received two evaluations of your Stage 2 submission. As you will see, both reviewers recognise the value of the study and appreciate the work involved in completing the registered protocol. I've now considered their comments alongside my own assessment and I'm pleased to invite a revision.

We thank the Recommender and reviewers for recognising the value of the study and our efforts to follow the registered protocol.

Regarding criterion 2A, Reeder raises concerns about the adequacy of the sample size, noting the number of inconclusive Bayes factors and the ~60-participant difference between the planned and

final analysed samples. However, I don't consider this a failure to meet the approved sampling plan. The Stage 1 manuscript specifies in Table 1 (p7) that "The goal is to recruit around 500 adult English speakers using Prolific", while the Participants section (p8) states "We aim to recruit around 500". You recruited the planned 500 participants, and neither statement requires a final sample of 500 after exclusions. In addition, the manipulation checks provide strong evidence that the illusions functioned as intended, and inconclusive Bayes factors don't necessarily indicate a failure to satisfy the approved outcome-neutral criteria. Overall, I therefore considered 2A met, although the implications of the exclusions for the strength of the evidence should be considered when interpreting the findings.

We thank the Recommender for considering criterion 2A met. We now consider the implications of the exclusions for the strength of the evidence when interpreting the findings. The fully registered analysis (N = 295) is reported as the main analysis, and the reduced sensitivity resulting from the exclusions is discussed as a limitation. An unregistered sensitivity analysis with a larger sample (N = 458 per illusion) led to the same conclusions for the primary outcome (Table A1 and A2).

Lines 536-546:
“But while our chosen summary metrics provided a straightforward, reproducible index of performance, collapsing the variance across task difficulty, illusion strength, and speed-accuracy trade-offs might have stripped away the nuanced resolution required to detect the subtle, low-level individual differences associated with personality-level characteristics. **The sensitivity of the tests was further limited by the sample size of the registered analysis: applying the registered space bar check reduced the sample from 468 to 295 participants, and the inconclusive results (concentrated in the Vertical-Horizontal illusion and the secondary outcome) may partly reflect this reduced sensitivity rather than evidence for a relationship. Notably, the unregistered analysis, with a larger sample, led to the same conclusions for the primary outcome.**”

However, several issues concerning adherence to the registered protocol, transparency of reporting, and interpretation of the findings need to be addressed before we can proceed to a positive Stage 2 recommendation. In addition to the reviewers' detailed comments, I'd also like to draw your attention to the following points:

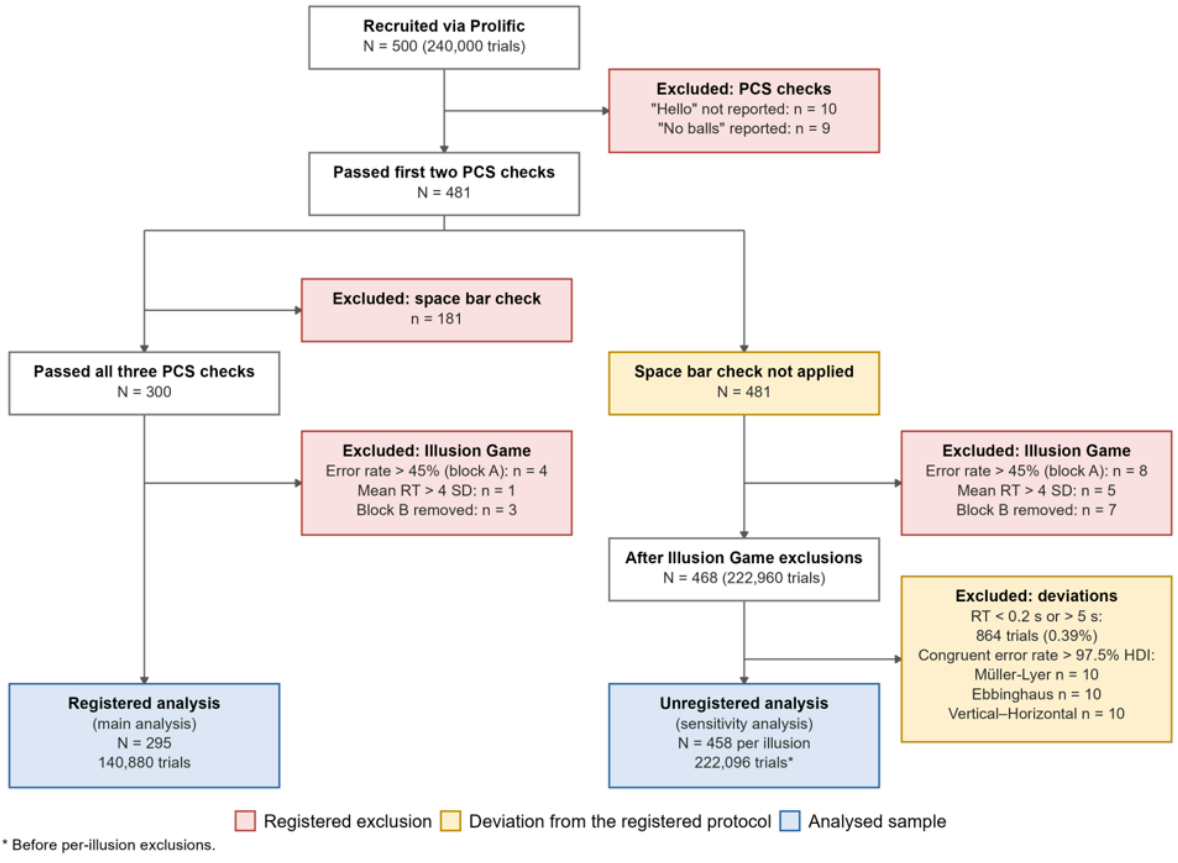
1. Both reviewers raise questions about the registered exclusion criteria. The Stage 1 protocol states (p13) that "any participant exhibiting an error rate greater than 45% for any illusion type will be considered to be responding at chance level", and specifies that participants exceeding this threshold in the first block will be excluded entirely, whereas only the second block will be removed if the criterion is met exclusively there. I can see that this criterion was implemented in your public cleaning script, specifically in analysis/1_cleaning.qmd, lines 181–193, where participants are flagged using p_Error > 0.45 | z_RT > 4. However, unless I missed it, the Stage 2 manuscript doesn't clearly report the application of the error-rate criterion, and the script combines error-rate and RT exclusions when determining whether to remove entire participants or individual blocks. At Stage 1 (p13), the block-specific rule was explicitly described for error rates, whereas the RT criterion was stated separately: "participants whose RTs are significantly slower than the group average (RT > 4 SD above the mean, based on Makowski et al., 2023) will be excluded from the analysis." Please clarify whether the approved rules were followed for each exclusion type and explain any differences. I'd also like a complete account of the exclusions, preferably in a CONSORT-style flow diagram, showing the number of participants and trials excluded at each stage, with the resulting sample sizes for each analysis. This should also resolve the apparent discrepancy between the sample sizes reported in the manuscript and on GitHub.

We thank the Recommender for this careful check. The Recommender is correct: our cleaning script combined the registered error-rate criterion (> 45%) and the RT criterion (> 4 SD) in a single flag and applied the block-specific rule to both, and the Stage 2 manuscript then described the resulting counts as if they arose from the RT criterion alone.

We have now separated the two criteria so that each follows the approved protocol. The error-rate criterion is applied with the registered block-specific rule, and the RT criterion at the participant level, as registered. In the fully registered analysis, which is now the main analysis, 4 of the 300 participants who passed all three PCS attention checks exceeded 45% in block A and were fully excluded, 3 exceeded it only in block B and had that block removed, and 1 participant whose mean RT exceeded the group mean by more than 4 SD was fully excluded, leaving 295 participants.

In the sample used for the unregistered sensitivity analysis (where the space bar check was not applied; see Comment 4), the same criteria fully excluded 8 participants on the basis of error rate and 5 on the basis of RT (none of whom were flagged by both criteria), and removed the second block of 7 participants, leaving 468 participants. Results under the block-wise RT rule used in the initial Stage 2 submission are reported in the analysis report (Sensitivity Analyses); applying it did not change the results for any outcome.

A CONSORT-style flow diagram (Figure A1) now reports the number of participants and trials excluded at each stage and the resulting sample size for each analysis.



The discrepancy with GitHub arose because the online report gave the sample size before the per-illusion exclusion discussed in Comment 2, whereas the manuscript gave it after. The manuscript and the online reports now report the same samples: 295 participants in the fully registered (main)

analysis, and 468 participants (458 per illusion after the per-illusion exclusion) in the unregistered sensitivity analysis.

Lines 153-157:
"**Following the exclusion steps (see Results and Figure A1), 295 participants (Mean age = 38.1, SD = 12.9, 48.8% women) were retained in the registered analysis. Sensitivity analyses addressing deviations from the registered protocol were conducted on a larger sample of 468 participants (Mean age = 38.7, SD = 13.2, 48.7% women), of whom 458 were analysed for each illusion (see Deviations From Registration).**"

Lines 305-314:
"**Of the 500 recruited participants, 19 failed the first two PCS attention checks (10 did not report hearing "Hello"; 9 reported that no balls were presented), and 189 failed the space bar check, 181 of whom had passed the first two checks (see Deviations From Registration). Of the 300 participants who passed all three checks, in line with the registered protocol, 4 participants with an error rate above 45% for any illusion type in the first block were fully excluded, 3 participants exceeding this threshold only in the second block had that block removed (these participants were retained), and 1 participant whose mean RT exceeded the group mean by more than 4 SD was fully excluded. This left 295 participants for the fully registered analysis. The full sequence of exclusions is shown in Figure A1.**"

2. The additional exclusion of participants whose error rates fell outside the 97.5% interval wasn't specified at Stage 1. Although the rationale for removing participants who appear to have misunderstood task instructions is reasonable, the implemented procedure warrants closer scrutiny. In particular, analysis/2_analysis.qmd, lines 145–180, under Illusion Game Scores → Outlier Removal → Exclusion, shows that the filtering is applied to both congruent and incongruent error-rate indices, using bayestestR::hdi(..., ci = 0.975), and flags observations outside either boundary or with missing values. This seems broader than the explanation in the manuscript, which emphasises unusually high errors in the congruent condition. Please clarify precisely how this criterion was implemented, explain the choice of the 97.5% threshold, and establish whether the procedure might have excluded participants with genuinely extreme illusion sensitivity rather than invalid responses. Please also clarify whether the reported interval should be described as a highest-density interval rather than a confidence interval, and provide a sensitivity analysis showing how the main findings change when these additional exclusions aren't applied.

We thank the Recommender for this scrutiny, which identified a genuine problem. As implemented in the initial Stage 2 submission, the procedure flagged participants outside the 97.5% interval of both the congruent and the incongruent error rates, in both tails, as well as participants with missing values. This was broader than its stated rationale: it also excluded participants with very low incongruent error rates, who plausibly reflect genuinely low illusion sensitivity rather than invalid responses. The interval is a highest density interval (HDI), computed with bayestestR::hdi(), not a confidence interval; this is now stated in the manuscript and in the analysis scripts.

We have made three changes:

(1) The HDI-based exclusion is not applied in the fully registered analysis, which is now the main analysis.

(2) In the unregistered sensitivity analysis, the procedure is restricted to its stated rationale: the upper tail of the congruent error rate, computed separately for each illusion after the removal of extreme trials. In the congruent condition the illusion makes the correct response easier, so errors are typically

rare, and highly susceptible participants would be expected to make fewer congruent errors, not more. A very high congruent error rate therefore cannot reflect extreme illusion sensitivity, and is more consistent with a misunderstanding of the instructions (e.g., selecting the smaller instead of the larger circle). The incongruent error rates, which index illusion sensitivity, are no longer used in this exclusion, and missing values are no longer flagged. This version excludes 10 participants (2.14%) per illusion, each from the analyses of that illusion only.

(3) The 97.5% level was chosen to flag only the most extreme responses. As it was not specified in advance, we do not rely on it: it is applied only in the sensitivity analysis.

As requested, Tables A1 and A2 compare the results of the unregistered analysis with and without this exclusion (N = 468 instead of 458 per illusion). Removing it did not change the classification of any test (Table A1), and changed the Bayes factors only slightly (Table A2).

Lines 356-364:
"**Second, during data inspection, a small number of participants showed very high error rates in the congruent condition, where the illusion makes the task easier and errors are typically rare. These patterns are more consistent with a misunderstanding of the instructions (e.g., selecting the smaller instead of the larger circle) than with illusion sensitivity, as greater susceptibility to the illusion should lead to fewer, not more, errors in this condition. Participants whose congruent error rate for a given illusion lay above the 97.5% highest density interval (HDI) of its distribution (computed after the removal of these trials) were therefore excluded from the analyses of that illusion only (10 participants, 2.14%, for each illusion), leaving 458 participants per illusion.**"

3. The removal of trials with reaction times <0.2 sec or >5 sec also appears to be an unregistered deviation. This appears to be implemented in analysis/1_cleaning.qmd, lines 258–268, under Illusion Game Trials. Although the Stage 1 protocol refers generally to removing "problematic participants and trials" (p13), it doesn't specify these trial-level RT thresholds. This is a relatively conventional preprocessing step and affected only 0.31% of trials, so I don't see it as inherently problematic. Nevertheless, please explicitly identify the additional thresholds as a deviation and report whether retaining these trials materially affects the main results.

We agree with this comment; the trial-level RT thresholds are now explicitly identified as a deviation from the Stage 1 protocol. The fully registered (main) analysis does not apply them; they are applied only in the unregistered sensitivity analysis, where they removed 864 trials (0.39%; the percentage differs slightly from the initial Stage 2 submission because it is now computed on the sample of the unregistered analysis).

Retaining these trials did not change the results for the primary outcome (error rate) or for the relationship between PCS and IES. For psychoticism and IES, two tests that provided evidence for the null became inconclusive, and none provided evidence for a relationship (Table A1 and A2).

Lines 351-355:
"**Two unregistered exclusions were applied to the Illusion Game data in the unregistered analysis. First, the Stage 1 protocol referred to the removal of problematic trials but did not specify a criterion. Trials with response times below 0.2 s or above 5 s were therefore excluded as implausible (Thériault et al., 2024), removing 864 trials (0.39% of the data; 298 faster than 0.2 s and 566 slower than 5 s).**"

4. I understand the decision not to apply the registered spacebar attention check, given that it would have excluded 189 participants. This seems a potentially defensible deviation, particularly if the check wasn't functioning as intended. However, the approved Stage 1 protocol does state (p13) that "Participants who fail any one of these will be excluded from further analysis." The fact that PCS internal consistency was unchanged by excluding these participants doesn't necessarily establish that the check was invalid. Please explain more precisely why you believe this check produced so many failures, and ensure that the results obtained with and without these participants are reported transparently. I think the relevant implementation is in analysis/1_cleaning.qmd, lines 119–135, under Exclusion → PCS, where outliers_pcs3 identifies participants failing the spacebar check but outliers_pcs includes only the first two checks. Importantly, the subsequent analysis excluding the spacebar-check failures should be distinguished from a fully registered analysis, because other unregistered preprocessing decisions appear to have been retained.

We thank the Recommender for pressing this point. Below, we (1) explain why we believe the check produced so many failures, (2) report the results with and without these participants, and (3) distinguish the analysis without the space bar check from the fully registered analysis.

(1) Reasons for the failure rate. Of the 189 participants who failed the check, 177 did not press the space bar at all, which argues against dropped key presses (which would produce partial counts). Most of them (181 of 189) passed the other two attention checks, and in the sample used for the sensitivity analyses (N = 468), their scores on the remaining PCS items were not lower than those of participants who passed (M = 2.49 vs. 2.28; BF_{10} = 2.90), suggesting that failures did not reflect general inattention. We believe instead that the failure rate reflects how the check was implemented. The auditory instruction, delivered with eyes closed, stated that the "PLEASE WAIT" message would appear "a little later". In the online implementation, however, the screen appeared as soon as the recording ended (i.e., as participants were told to open their eyes), disappeared after 10 seconds, and only recorded presses made while it was displayed. Participants who were still reorienting after opening their eyes, or who pressed while the instruction was still playing, would therefore have been recorded as not pressing at all. We have removed the argument based on internal consistency, which, as the Recommender notes, did not establish that the check was invalid.

(2) Results with and without these participants. Tables A1 and A2 compare the fully registered analysis, in which these participants are excluded (N = 295), with an analysis that applies all registered criteria except the space bar check, with no other deviation (N = 468); Table A2 reports the Bayes factor of every test in both. Excluding or retaining these participants led to the same conclusions for the primary outcome (error rate). For the secondary outcome (IES), the classification of three tests differed when these participants were retained: the relationships of PCS and of psychoticism with IES for the Vertical-Horizontal illusion (Mild) were inconclusive rather than supporting the null (PCS: BF_{10} = 0.38 vs. 0.32; psychoticism: BF_{10} = 0.74 vs. 0.14), and the relationship between psychoticism and IES for the Vertical-Horizontal illusion (Strong) provided evidence for a relationship (BF_{10} = 3.21 vs. 0.92), although in the direction opposite to the registered hypothesis.

(3) Distinction from the fully registered analysis. Because the check was registered, the main analysis is now the fully registered analysis, in which all three checks are applied and no unregistered preprocessing step is used. The analysis without the space bar check described in (2) differs from it only in this check. The unregistered analysis, which combines all deviations (N = 458 per illusion), is reported separately as a sensitivity analysis.

Lines 331-350:
"The registered protocol specified that participants failing any of **three** attention checks in the PCS would be excluded. **The first two checks were failed by 19 participants (3.8%), whereas**

the third one was failed by 189 participants (37.8%). Most failures (177 of 189) recorded no presses at all, which argues against dropped key presses (which would produce partial counts). Most of these participants (181 of 189) passed the other two checks, and in the sample used for the sensitivity analyses (N = 468), the scores of participants who failed the space bar check on the remaining PCS items were not lower than those of participants who passed it (M = 2.49 vs. 2.28; BF_{10} = 2.90), suggesting that failures did not reflect general inattention. Instead, the failure rate may reflect how the check was implemented: although the auditory instruction, delivered with eyes closed, stated that the "PLEASE WAIT" message would appear "a little later", the screen appeared as soon as the recording ended (i.e., as participants were told to open their eyes), disappeared after 10 seconds, and only recorded presses made while it was displayed. Participants who were still reorienting after opening their eyes, or who pressed during the instruction, would therefore have been recorded as not pressing at all. We therefore repeated the analyses without the space bar check as part of the unregistered analysis. In this sample (the 481 participants who passed the first two PCS checks), the registered Illusion Game criteria fully excluded 8 participants on the basis of error rate and 5 on the basis of RT (none of whom were flagged by both criteria), and removed the second block of 7 participants (who were retained), leaving 468 participants."

Lines 433-439:
"**Applying the registered criteria without the space bar check (N = 468) led to the same conclusions for the primary outcome. For the secondary outcome (IES), two tests of the Vertical-Horizontal illusion (Mild) were inconclusive rather than supporting the null (PCS: BF_{10} = 0.38; psychoticism: BF_{10} = 0.74), and one test of the relationship between psychoticism and IES provided evidence for a relationship (Vertical-Horizontal, Strong: BF_{10} = 3.21; Table A2), although in the direction opposite to the registered hypothesis.** "

5. The approved Stage 1 protocol specified a conditional decision rule for combining Mild and Strong illusion-strength conditions. Specifically, it states (p14): "If these correlations are sufficiently high (r > .50, Cohen, 2013), the groups will be collapsed and outcomes recomputed across all relevant trials", followed by "If not, the groups will be analysed separately." At Stage 2, you report that some correlations exceeded this threshold and others did not, but then elected to analyse all strength conditions separately. This appears to depart from the registered decision rule. The relevant correlations are computed in analysis/2_analysis.qmd, lines 208–263, under Should we collapse across Mild and Strong, and Easy and Hard? → Mild vs. Strong. Please report the relevant correlations, explain the change, and present the results obtained by applying the original conditional rule. Any alternative approach can then be retained as an explicitly labelled supplementary or exploratory analysis.

We thank the Recommender for this comment. We now apply the registered conditional rule as specified, separately for each illusion and outcome and separately for each analysis. The correlations between the Mild and Strong conditions are reported in Table A3; the decision is based on the sample correlation, so that it is deterministic (the Bayesian estimates are virtually identical).

In the fully registered (main) analysis, all three error-rate correlations exceeded .50 (Müller-Lyer: r = .67; Ebbinghaus: r = .76; Vertical–Horizontal: r = .72), so error rates were collapsed across Mild and Strong. For IES, the Müller-Lyer (r = .55) and Ebbinghaus (r = .54) conditions were collapsed, and the Vertical–Horizontal conditions (r = .43) were analysed separately. In the unregistered sensitivity analysis, the rule was applied anew and led to the same decisions (error rates: r = .66, .72 and .74; IES Ebbinghaus: r = .58; IES Vertical–Horizontal: r = .40), except for the IES of the Müller-Lyer illusion (r = .43), which was analysed separately.

Analysing all strength conditions separately, as in the initial Stage 2 submission, is retained as an explicitly labelled exploratory analysis (Exploratory Analyses; Table A1, last row).

Lines 381- 388:
"**For the illusion scores, the registered plan was to first evaluate whether the Mild and Strong illusion strength conditions could be collapsed, based on the correlation between them for each illusion and outcome (collapsed if r > .50; Table A3). Error rates were collapsed for all three illusions (Müller-Lyer: r = .67; Ebbinghaus: r = .76; Vertical–Horizontal: r = .72). IES were collapsed for the Müller-Lyer (r = .55) and Ebbinghaus illusions (r = .54), but analysed separately for the Vertical–Horizontal illusion (r = .43). Analyses with Mild and Strong kept separate for all scores are reported as exploratory analyses.**"

Lines 429-432:
"**The registered decision rule for collapsing Mild and Strong was applied separately in this sample, leading to the same decisions except for the IES of the Müller-Lyer illusion, which was analysed separately (r = .43; Table A3)**."

6. I identified a potentially important discrepancy between the reported results and Figure 2 when examining the analysis scripts. In analysis/2_analysis.qmd, lines 495–528, under Correlations → All, the function make_cor() assigns the correlation results to a global object called rez. However, the subsequent call under Correlations → PCS3 Excluded (lines 532–535) appears to overwrite this object with results from the sensitivity analysis excluding participants who failed the spacebar check. Figure 2 is then generated from p <- rez under Figures (lines 541–554). As a result, Figure 2 appears to display the sensitivity analysis rather than the analysis retaining these participants that you describe as primary. For example, the correlation between PCS and Vertical-Horizontal error rate under Mild illusion strength yields BF_{10} = 3.03 in the retained-participant analysis but BF_{10} = 2.61 after excluding the spacebar-check failures. More generally, across the 12 Mild/Strong PCS correlations, I believe 7 meet the registered threshold for evidence favouring the null in the retained-participant analysis, compared with 10 in the sensitivity analysis. Please check this carefully, ensure that the figures and numerical results correspond to their intended analyses, and clarify any differences between the primary and sensitivity analyses. The claim that excluding the spacebar-check failures had no impact on the findings also needs to be qualified in light of these differences.

We are very grateful to the Recommender for identifying this error. The function stored its results in a global object, which the subsequent sensitivity analysis overwrote, so Figure 2 displayed the analysis excluding the space bar check failures rather than the analysis described as primary. The function now returns its results, every table and figure is generated from an explicitly named analysis, and we have checked all reported values against the regenerated output.

We have also removed the claim that excluding these participants "did not have an impact", and we confirm the Recommender's count: in the initial Stage 2 submission's Mild/Strong breakdown, 7 of the 12 PCS correlations met $BF_{10} \leq 1/3$ when these participants were retained, compared with 10 of 12 when they were excluded. In the revised manuscript, the main analysis is the fully registered analysis (N = 295; see Comment 4), which is the analysis shown in Table 3 and Figure 2. The effect of excluding the space bar check failures is shown in Tables A1 and A2, which compare this analysis with one that applies all registered criteria except the space bar check (N = 468). The conclusions for the primary outcome (error rate) were the same in both. For the secondary outcome (IES), the classification of three tests differed when these participants were retained: the relationships of PCS and of psychoticism with IES for the Vertical-Horizontal illusion (Mild) were inconclusive rather than supporting the null (PCS: BF_{10} = 0.38 vs. 0.32; psychoticism: BF_{10} = 0.74 vs. 0.14), and the

relationship between psychoticism and IES for the Vertical-Horizontal illusion (Strong) provided evidence for a relationship (BF_{10} = 3.21 vs. 0.92), although in the direction opposite to the registered hypothesis.

Lines 433–439:
"**Applying the registered criteria without the space bar check (N = 468) led to the same conclusions for the primary outcome. For the secondary outcome (IES), two tests of the Vertical-Horizontal illusion (Mild) were inconclusive rather than supporting the null (PCS: BF_{10} = 0.38; psychoticism: BF_{10} = 0.74), and one test of the relationship between psychoticism and IES provided evidence for a relationship (Vertical-Horizontal, Strong: BF_{10} = 3.21; Table A2), although in the direction opposite to the registered hypothesis.** "

7. I agree with both reviewers that the main findings need to be substantially easier to evaluate. Table 1 of the approved Stage 1 manuscript specifies (p7): "For the PC-VI sensitivity relationship, we will interpret $BF_{10} \leq 1/3$ as evidence against a relationship". Please include a complete table of the registered correlations, reporting the correlation estimates, Bayes factors, and relevant sample sizes, and explicitly identify how many analyses met the prespecified evidential thresholds and how many were inconclusive. The manuscript should also report descriptive statistics and distributions for PCS and psychoticism. I note that the approved protocol identifies error rate as the "primary outcome" and IES as a "secondary exploratory outcome" (p12), so please preserve this distinction when presenting the results. The complete analysis reports on GitHub are valuable for reproducibility, but they shouldn't substitute for accessible reporting of the main findings in the article or its supplementary materials.

We thank the Recommender and both reviewers for this suggestion. The main findings are now reported in full in the article, so that they can be evaluated without the online reports:

(1) Complete table of registered correlations. Table 3 reports all 14 registered tests of the fully registered (main) analysis (N = 295), with the posterior median correlation and 95% credible interval, the Bayes factor, and the evidence category according to the prespecified thresholds ($BF_{10} \leq 1/3$: evidence for the null; $BF_{10} \geq 3$: evidence for a relationship). Figure 2 displays these tests together with the separate Mild/Strong and Congruent.

(2) Number of tests meeting the thresholds. The Results now state that 10 of the 14 registered tests met the threshold for evidence for the null, 4 were inconclusive, and none supported a relationship, and break these numbers down by trait and outcome. Table A1 gives the same counts for each trait and outcome under every version of the analysis, and Table A2 the Bayes factor of every registered test in each version.

(3) Primary vs. secondary outcome. Error rate is labelled as the primary outcome and IES as the secondary outcome throughout: in the text (which reports the primary outcome first), in Table 3 and in Figure 2, where they are shown in separate panels.

(4) Descriptive statistics and distributions. The Results now report the mean, SD and range of PCS and psychoticism scores, and the skewness of psychoticism, and their distributions are shown in Figure A2. Descriptive statistics for the illusion scores are given in Table 2.

Lines 377–380:

"**PCS scores (possible range 0–5) had a mean of 2.16 (SD = 0.80, range: 0.00–4.20), and psychoticism scores (possible range 0–3) a mean of 0.48 (SD = 0.40, range: 0.00–1.80), with a positively skewed distribution (skewness = 0.85; Figure A2).**"

Lines 395-408:

“Bayesian correlations were conducted to assess the relationship between illusion sensitivity **(error rate as the primary outcome and IES as the secondary outcome)** and both PCS scores and the psychoticism facet of the PID-5. **All registered tests are reported in Table 3 and displayed in Figure 2. Of the 14 registered tests, 10 met the threshold for evidence for the null ($BF_{10} \leq 1/3$), 4 were inconclusive, and none supported a relationship. For the primary outcome (error rate), the correlation between PCS and illusion sensitivity provided evidence for the null for the Müller-Lyer ($\rho = 0.04$, $BF_{10} = 0.19$) and Ebbinghaus illusions ($\rho = 0.02$, $BF_{10} = 0.14$), and was inconclusive for the Vertical-Horizontal illusion ($\rho = 0.08$, $BF_{10} = 0.38$). For the secondary outcome (IES), 3 of the 4 PCS tests supported the null and 1 was inconclusive (Müller-Lyer, $BF_{10} = 0.45$). The correlations between psychoticism and error rates all supported the null ($BF_{10} = 0.14–0.17$), as did 2 of the 4 IES tests, while the other 2 were inconclusive ($BF_{10} = 0.40$ and $BF_{10} = 0.92$).**”

One further technical point warrants clarification: the Stage 1 analysis plan specified BayesFactor::correlationBF() with a medium prior (r-scale = 1/3; Table 1, p7; also Methods, p14), whereas the correlation analyses in analysis/2_analysis.qmd, lines 495–502 use correlation::correlation(..., bayesian = TRUE) without explicitly specifying that prior. Please confirm that the implemented analysis uses the approved prior and produces equivalent Bayes factors (or explain and evaluate any deviation if there is one).

We thank the Recommender for this check. We confirm that the implemented analysis uses the approved prior. With bayesian = TRUE, correlation::correlation() computes Bayes factors with BayesFactor::correlationBF(), and its default prior (“medium”) corresponds to an r-scale of 1/3, as specified at Stage 1. The prior was therefore applied as registered in the initial Stage 2 submission, but it was not stated in the code. It is now specified explicitly (analysis/2_analysis.qmd, Correlations: bayesian_prior = “medium”), and we verified that computing the Bayes factors directly with BayesFactor::correlationBF(..., rscale = 1/3) yields identical values. There is therefore no deviation.

8. Both reviewers correctly identify instances where the conclusions are stronger than the evidence warrants. In the analysis retaining participants who failed the spacebar check, for example, only 7 of the 12 Mild/Strong PCS correlations satisfy the registered $BF_{10} \leq 1/3$ criterion for evidence favouring the null, while 4 are inconclusive and 1 favours an association. This pattern doesn't justify categorical statements that the absence of a relationship has been confirmed across visual illusions. I'd therefore encourage you to distinguish more carefully between evidence favouring the null and an absence of compelling evidence for an association, and to qualify broader claims about cognitive impenetrability.

We agree with the Recommender and with both reviewers. We have revised the manuscript throughout to distinguish evidence favouring the null ($BF_{10} \leq 1/3$) from inconclusive results, and to restrict our conclusions to the illusions and outcomes for which the evidence favoured the null.

In the fully registered (main) analysis, 10 of the 14 registered tests provided evidence for the null, 4 were inconclusive, and none supported a relationship (Table 3). For PCS and the primary outcome (error rate), the evidence favoured the null for the Müller-Lyer and Ebbinghaus illusions but was inconclusive for the Vertical-Horizontal illusion. When Mild and Strong were analysed separately (exploratory analysis), 9 of the 12 PCS correlations provided evidence for the null and 3 were inconclusive (Table A1). We no longer state that the absence of a relationship was confirmed across illusions. Accordingly, the claim of cognitive impenetrability is now limited to the Müller-Lyer and Ebbinghaus illusions under the present task conditions, and the Discussion explicitly states that it

cannot be extended to the Vertical-Horizontal illusion. We also note, as a limitation, that the reduced sample size of the registered analysis may have contributed to the inconclusive results.

Lines 469–474:
"**For the primary outcome (error rate), registered Bayesian correlation analyses provided evidence against a relationship between phenomenological control and sensitivity to two of the three illusions tested (Müller-Lyer and Ebbinghaus), whereas the evidence for the third (Vertical-Horizontal) was inconclusive, partly supporting the first hypothesis. For psychoticism, the data favoured the absence of an association with error rates, contrary to the second hypothesis, whereas the evidence for IES was mixed.**"

Lines 483-493:
"This **pattern is consistent with the cognitive impenetrability of these illusions under the present task conditions. […] The inconclusive result for the Vertical-Horizontal illusion, however, indicates that this conclusion cannot be extended to all three illusions tested.**"

Lines 540-546:
"**The sensitivity of the tests was further limited by the sample size of the registered analysis: applying the registered space bar check reduced the sample from 468 to 295 participants, and the inconclusive results (concentrated in the Vertical-Horizontal illusion and the secondary outcome) may partly reflect this reduced sensitivity rather than evidence for a relationship. Notably, the unregistered analysis, with a larger sample, led to the same conclusions for the primary outcome.** "

Abstract:
"**[…] These findings are consistent with the cognitive impenetrability of these two illusions under the present task conditions, and suggest that susceptibility to them is not related to trait-level control over subjective experience.**"

> I also agree with Palfi's observation that motivating participants to overcome an illusion isn't equivalent to providing them with strategies for doing so. The distinction between trait phenomenological control, state-dependent interventions, and actual behavioural control over illusion susceptibility deserves clearer treatment.

We also agree with Dr Palfi's observation. The Discussion now distinguishes more clearly between trait phenomenological control, which was measured in this study, state-dependent interventions such as suggestions or strategies, which were not used, and actual behavioural control over illusion susceptibility. In particular, we now note that our instructions motivated participants to resist the illusion but did not provide a strategy for doing so, and we limit our conclusions accordingly: the results concern trait phenomenological control and do not speak to whether state-dependent interventions could confer behavioural control over these illusions. We also suggest that future studies test whether explicit strategies, such as those shown to reduce Stroop interference (Palfi et al., 2022), reduce susceptibility to these illusions (see also our response to Dr Palfi, Interpretation and limitations section).

Lines 555-567:
"Furthermore, the present study measured trait phenomenological control, the baseline capacity for altering subjective experience, but did not provide an explicit induction or suggestion to alter the perception of the stimuli (i.e., state phenomenological control). **Moreover, while the task instructions provided a clear goal (to respond accurately despite the illusion), they did not provide any strategy for achieving it, and motivation to overcome an illusion is distinct from knowing how to do so. The present results therefore concern the relationship between trait**

phenomenological control and illusion susceptibility, and do not speak to whether state-dependent interventions (e.g., a specific suggestion or strategy) could confer behavioural control over these illusions. Future studies could test whether explicitly providing potentially effective strategies, such as those shown to reduce Stroop interference (Palfi et al., 2022), reduces susceptibility to these illusions, and whether the present instructions are sufficient to reduce interference in tasks in which cognitive penetration is known to be possible."

> The low consistency between some illusion measures also needs to inform the scope of your conclusions.

We agree with the Recommender. The low consistency between the illusion measures (error rate: $\alpha = .19$ for the collapsed scores) indicates that performance on each illusion is largely driven by illusion-specific mechanisms, and that the three illusions cannot be treated as interchangeable indices of a general illusion sensitivity. We have therefore limited the scope of our conclusions: they are now stated separately for each illusion, apply only to the illusions for which evidence for the null was obtained (Müller-Lyer and Ebbinghaus), and are explicitly not generalised to visual illusions in general.

Lines 576-582:
"**In summary, the present study suggests that sensitivity to the Müller-Lyer and Ebbinghaus illusions is largely unrelated to our trait-level capacity to alter subjective phenomenology. Importantly, three illusions are not a representative sample of visual illusions, and their low inter-correlation in the present data suggests that performance on each of them is largely driven by illusion-specific mechanisms. Our conclusions therefore apply to the illusions for which evidence for the null was obtained, rather than to visual illusions in general.**"

9. Please address the reviewers' remaining comments, including the clarity and labelling of Figure 2, the English-language eligibility requirements, and the change in the reported duration of the original Illusion Game experiment. On this latter point, the Stage 1 manuscript states (p10) that the original experiment lasted "approximately 55 minutes", whereas the Stage 2 manuscript gives 48 minutes. Please check which figure is correct.

Each of these points is addressed in detail in our responses to the reviewers; we summarise the changes here.

Figure 2. The figure has been redrawn (see our responses to Dr Reeder's comment on Figure 2 and Dr Palfi's comment on the readability of the results). It now displays the fully registered (main) analysis, which corrects the error identified in Comment 6. The primary (error rate) and secondary (IES) outcomes are shown in separate, labelled panels, and conditions in which Mild and Strong were collapsed are labelled "Mild + Strong". Bayes factors are reported as values, with symbols marking the prespecified thresholds, and the Congruent condition, which does not index illusion sensitivity, is marked as shown for reference only. The caption explains all labels and symbols.

Duration of the original Illusion Game experiment. The Stage 1 figure (approximately 55 minutes) is correct. The 48 minutes given in the initial Stage 2 submission was the median completion time of the present study (47.96 minutes), which was inserted into the description of the original experiment by mistake. This has been corrected.

Lines 188-189:
"Participants completed a total of 1,340 trials, with the experiment lasting approximately **55** minutes."

> The Stage 1 Participants section also specifies "adult English native speakers" (p8), whereas the Stage 2 manuscript describes English-speaking adults, so please clarify whether this reflects a change in eligibility or simply a correction to the description.

We thank the Recommender for raising this point. This reflects a correction to the description rather than a change in eligibility during the study: participants were recruited with Prolific's screening filter for participants whose first or primary language is English, which was the intended criterion throughout, and the filter was not changed at any point. The Stage 1 description ("adult English native speakers") did not reflect this criterion and was an error in the Stage 1 manuscript. Consistent with this, the approved study design table (Table 1) specified "adult English speakers". Because the recruited sample nevertheless differs from the approved description, we report this as a deviation from the Stage 1 protocol. Based on participants' Prolific demographic data, 59 of the 500 recruited participants (43 of the 295 participants in the registered analysis) did not report English as their first language. Restricting the registered analysis to native English speakers (N = 252), as specified in the Stage 1 protocol, did not change the conclusions for the primary outcome (Table A1 and Table A2).

Lines 323–330:
"**The Stage 1 manuscript specified that adult native English speakers would be recruited. Eligibility was implemented using Prolific's screening filters, which admitted participants whose first or primary language was English; the Stage 1 wording did not reflect this intended recruitment criterion and was an error in the Stage 1 manuscript. As a result, 59 of the 500 recruited participants (43 of the 295 participants in the registered analysis) did not report English as their first language. The effect of this deviation was evaluated by restricting the registered analysis to native English speakers (N = 252; see Sensitivity Analyses and Table A1 and Table A2).**"

> Similarly, please confirm that informed consent was obtained as approved, given the change in wording between the two manuscripts.

We confirm that informed consent was obtained as approved. Participants were presented with an information sheet and consent form at the start of the experiment and could only proceed by clicking "I read, understood, and I consent". The change in wording was unintentional, and the Stage 1 wording has been restored (changed only to the past tense).

Lines 149-151:
"Participants were first presented with an explanatory statement and **the consent form**, and could proceed by pressing a button to confirm they had read and understood the information."

> The revised manuscript should consistently describe completed procedures in the past tense, avoid NHST terminology when interpreting Bayes factors, and clearly distinguish registered from exploratory analyses.

We have revised the Methods so that all completed procedures and analyses are described in the past tense. We also replaced all NHST terminology (e.g., "significant", "significantly") with Bayes factor-based descriptions of the evidence, and labelled each analysis as registered, a deviation from the registered protocol (with its sensitivity analysis), or exploratory.

> Finally, references in the Results and Discussion to a hypothesised positive association between psychoticism and illusion sensitivity should be corrected to negative. The approved Stage 1 hypothesis explicitly states (Table 1, p7) that "higher psychoticism scores will be negatively associated with VI sensitivity". As you know, this direction was clarified and

corrected during Stage 1 review (via email), just before IPA was awarded, so there's no concern here about an unauthorised change to the registered hypothesis.

We thank the Recommender for this correction and for confirming the history of the hypothesis. All references to the hypothesised association between psychoticism and illusion sensitivity now describe it as negative, in line with the approved Stage 1 hypothesis (Table 1). The only remaining use of "positive" describes the observed direction of one exploratory correlation, which is explicitly noted as opposite to the registered hypothesis.

Line 287-289:
"Based on prior research (Makowski et al., 2023), we **expected** to find evidence ($BF_{10} \geq 3$) supporting a **(negative)** relationship between the psychoticism facet of the PID-5 and illusion sensitivity."

Lines 408-409:
"**None of the registered tests therefore supported the hypothesised negative association between psychoticism and illusion sensitivity.**"

Line 466-469:
"This study aimed to replicate previous findings indicating (1) an absence of a relationship between visual illusion sensitivity and phenomenological control (Lush et al., 2022), and (2) the existence of a **negative** relationship between illusion sensitivity and psychoticism (Makowski et al., 2023)."

Finally - on a general note, I'd encourage you to organise the revised Results section so that readers can follow the progression from registered preprocessing and analyses, through any deviations and their sensitivity analyses, to the exploratory findings. Ideally, the results obtained under the approved protocol should be clearly identifiable, with subsequent changes evaluated in a way that makes their consequences transparent for the reader.

We thank the Recommender for this suggestion, which we have followed in restructuring the Results. The section now progresses from the registered protocol, through the deviations and their consequences, to the exploratory findings:

(1) **Data Exclusion** → Registered Exclusions applies the exclusion criteria of the approved protocol, yielding the sample of the fully registered analysis (N = 295; Figure A1).

(2) **Data Exclusion** → Deviations From Registration describes each deviation from the Stage 1 protocol and its rationale (the English-language eligibility criterion, the space bar check, and the trial- and participant-level exclusions).

(3) **Manipulation Check and Score Consistency**, including the registered decision rule for collapsing Mild and Strong, are reported for the registered sample.

(4) **Correlations** → Registered Analysis reports the results obtained under the approved protocol (Table 3, Figure 2), which form the main analysis.

(5) **Correlations** → Sensitivity Analyses evaluates the consequences of the deviations, both combined (the unregistered analysis, N = 458 per illusion) and one at a time (Table A1), so that the effect of each change on the conclusions is transparent.

(6) **Correlations** → Exploratory Analyses reports the analyses that were not part of the registered protocol (Mild and Strong analysed separately; Easy and Hard trials).

Review by Bence Palfi

I would like to thank the authors for completing this interesting study and for following through on the research registered at Stage 1. Overall, I found the work valuable, but I have several comments concerning adherence to the Stage 1 protocol, reporting of the registered analyses, and the interpretation of the findings.

We thank Dr Palfi for his positive evaluation of our work and for his constructive comments, which we address below.

Exclusion criteria

I identified some deviations from the Stage 1 exclusion criteria that should be clarified and justified.

- I could not identify the registered filtering based on error rates in the Stage 2 manuscript. Were there no participants who met the preregistered criterion?

We thank Dr Palfi for this comment. The registered criterion was applied, but the initial Stage 2 submission did not report it separately. In the fully registered (main) analysis, 4 participants exceeded the 45% error-rate threshold in the first block and were fully excluded, and 3 exceeded it only in the second block and had that block removed (8 and 7, respectively, in the sample used for the unregistered sensitivity analysis). These exclusions are now reported explicitly, together with all other exclusions, in the Results and in a flow diagram (Figure A1). Please see our response to the Recommender, Comment 1, for details.

- Instead, an additional exclusion criterion appears to have been applied: “participants outside the 97.5% CI for the error rate for a given illusion had their data for that specific illusion excluded, affecting 20 participants (4.31%) for the Müller–Lyer illusion, 19 (4.09%) for the Ebbinghaus illusion, and 18 (3.88%) for the Vertical–Horizontal illusion”. As I could not identify this criterion in the Stage 1 protocol, this deviation should be clearly reported and justified.

We agree that this criterion was not registered. As explained in our response to the Recommender (Comment 2), the procedure was also broader than its stated rationale, and the interval was a highest density interval (HDI) rather than a confidence interval. The exclusion is not applied in the fully registered (main) analysis. It is now reported as a deviation and is applied only in the unregistered sensitivity analysis, restricted to its rationale: very high error rates in the congruent condition, where the illusion makes the task easier, indicate a misunderstanding of the instructions rather than illusion sensitivity. This version excludes 10 participants per illusion. Removing it did not change the results (Table A1 and A2).

- There also appears to have been a change in how RT-based outliers were excluded. My reading of the Stage 1 protocol is that the block-based exclusion rule was specified for error-rate filtering rather than RT-based filtering. Please clarify this discrepancy.

We agree with this comment. The block-based rule was registered for the error-rate criterion only, and the initial Stage 2 submission applied it to both criteria. The two criteria are now separated, and the RT criterion is applied at the participant level, as registered: in the fully registered (main) analysis, 1 participant whose mean RT exceeded the group mean by more than 4 SD was fully excluded (5 in the sample used for the unregistered sensitivity analysis). Results under the block-wise RT rule used in the initial Stage 2 submission are reported in the analysis report and did not change the results for any outcome. Please see our response to the Recommender, Comment 1, for details.

- Additional trials were excluded on the basis of very short or long RTs. I could not identify this exclusion criterion in the IPA version. If this was not preregistered, it should be explicitly reported as a deviation and how it impacted the results.

We thank Dr Palfi for this comment. This criterion was not preregistered: the Stage 1 protocol referred to the removal of problematic trials without specifying a criterion. It is now explicitly reported as a deviation and is not applied in the fully registered (main) analysis. In the unregistered sensitivity analysis, trials with response times below 0.2 s or above 5 s were excluded (864 trials, 0.39%). Retaining these trials did not change the results for the primary outcome (error rate); for the secondary outcome, two tests of the relationship between psychoticism and IES were inconclusive rather than supporting the null (Table A1 and A2). Please see our response to the Recommender, Comment 3, for details.

- A relatively large number of participants failed the first attention check. The manuscript describes this check as not being sufficiently well calibrated, but it is unclear what this means. Please explain the specific issue with the attention check and why ignoring it is a justified deviation from the registered procedure.

We agree with Dr Palfi's comment. The vague description ("not sufficiently well calibrated") has been removed, and the manuscript now explains the specific issue with this check. Of the 189 participants who failed it, 177 did not press the space bar at all, which argues against dropped key presses. Most of them (181 of 189) passed the other two attention checks, and their scores on the remaining PCS items were not lower than those of participants who passed (M = 2.49 vs. 2.28; BF_{10} = 2.90) in the sample used for the sensitivity analyses (N=468), suggesting that failures did not reflect general inattention. We believe instead that the failure rate reflects how the check was implemented: although the auditory instruction stated that the "PLEASE WAIT" message would appear "a little later", the screen appeared as soon as the recording ended (i.e., as participants were told to open their eyes), disappeared after 10 seconds, and only recorded presses made while it was displayed. Participants who were still reorienting after opening their eyes, or who pressed during the instruction, would therefore have been recorded as not pressing at all.

Because the check was registered, the main analysis is now the fully registered analysis, in which all three checks are applied (N = 295). Not applying the check is treated as a deviation and evaluated in a sensitivity analysis that applies all other registered criteria (N = 468). Both lead to the same conclusions for the primary outcome (error rate); for the secondary outcome, two tests of the Vertical-Horizontal illusion (Mild) became inconclusive, and one test of the relationship between psychoticism and IES (Vertical-Horizontal, Strong; BF_{10} = 3.21) provided evidence for a relationship when these participants were retained, although in the direction opposite to the registered hypothesis (Table A1 and A2). Please see our response to the Recommender, Comment 4, for details.

Statistical language

Please avoid NHST terminology, particularly the word “significant”, when interpreting Bayes factors.

All NHST terminology (e.g., “significant”, “significantly”) has been removed, and evidence is now described exclusively in terms of Bayes factors and the prespecified evidential thresholds (see also our response to the Recommender, Comment 9 and following).

Transparency of results

The reporting of the registered analyses should be more transparent. At minimum, the manuscript should state how many of the registered tests met the authors’ prespecified evidential threshold. Preferably, the results of all registered analyses should be reported in the manuscript or its supplementary materials rather than directing readers to a table in the GitHub repository.

We agree with this suggestion. All registered tests are now reported in the manuscript (Table 3), with their correlation estimates, 95% credible intervals, Bayes factors, sample size and evidential classification, and are displayed in Figure 2. The Results now state how many tests met each prespecified threshold: of the 14 registered tests, 10 provided evidence for the null ($BF_{10} \leq 1/3$), 4 were inconclusive, and none supported a relationship. The corresponding counts for every version of the analysis (sensitivity and exploratory analyses) are reported in the supplementary materials (Table A1 and A2), so that the article no longer relies on the GitHub repository for the main findings. Please see our response to the Recommender, Comment 7, for details.

A similar issue applies to some of the main results. Although these are presented in Figure 2, the values are difficult to read because of the small print. Providing the key numerical results in the text would make the findings substantially easier to evaluate.

We thank Dr Palfi for this suggestion. The key numerical results (correlation estimates and Bayes factors for the primary and secondary outcomes) are now given in the Results text, and all values are reported in Table 3 (see our response to the Recommender, Comment 7). Figure 2 has also been redesigned for readability: the primary (error rate) and secondary (IES) outcomes are shown in separate rows of panels with more spacing between them, the Bayes factors are placed at the right of each panel, clear of the reference line, and reported to two decimals, and the x-axis is labelled (see also our response to Dr Reeder’s comment on Figure 2).

Interpretation and limitations

The conclusions should avoid generalising the findings to visual illusions as a whole, given that clear evidence was obtained for only two of the three illusions examined. The conclusions should more closely reflect this variability across the tested illusions. More broadly, even if consistent evidence had been obtained across all three illusions, it is unclear to what extent findings from three specific illusions can be generalised to visual illusions as a whole. The manuscript would benefit from greater clarity about the intended scope of this generalisation and the justification for it.

We agree. The conclusions are now restricted to the illusions tested: the Discussion and abstract state that evidence for the null was obtained for the Müller-Lyer and Ebbinghaus illusions, and explicitly note that this conclusion cannot be extended to the Vertical–Horizontal illusion, for which the

evidence was inconclusive (see our response to the Recommender, Comment 8). We have also clarified the scope of the conclusions: three illusions are not a representative sample of visual illusions, and their low inter-correlation in our data suggests that performance on each of them is largely driven by illusion-specific mechanisms. Our conclusions are therefore limited to the illusions for which evidence for the null was obtained, and are not generalised to visual illusions as a class.

Lines 578-582:
"**Importantly, three illusions are not a representative sample of visual illusions, and their low inter-correlation in the present data suggests that performance on each of them is largely driven by illusion-specific mechanisms. Our conclusions therefore apply to the illusions for which evidence for the null was obtained, rather than to visual illusions in general.**"

The illusion game provides a very neat context for testing whether participants high in phenomenological control (PC) can reduce the influence of an illusion when explicitly encouraged to act in a goal-directed manner. However, motivation to overcome an illusion and knowledge of how to do so are potentially distinct. The instructions may provide a clear goal without necessarily giving participants sufficient information about strategies that could help them achieve it (see an example for reduing the Stroop effect with an imagination based strategy in Palfi et al., 2021, Cortex). This distinction would be useful to acknowledge as a limitation and direction for future research. A future study could test whether the current instructions are sufficient to reduce behavioural interference in contexts where cognitive penetration is possible, or whether explicitly providing potentially effective strategies further reduces susceptibility to these visual illusions.

We thank Dr Palfi for this insightful point. We have added it to the Discussion as a limitation and a direction for future research, together with a clearer distinction between trait phenomenological control, state-dependent interventions, and behavioural control over illusion susceptibility (see also our response to the Recommender, Comment 8). We cite the imagination-based strategy for reducing Stroop interference as Palfi et al. (2022).

Lines 558-567:
"**Moreover, while the task instructions provided a clear goal (to respond accurately despite the illusion), they did not provide any strategy for achieving it, and motivation to overcome an illusion is distinct from knowing how to do so. The present results therefore concern the relationship between trait phenomenological control and illusion susceptibility, and do not speak to whether state-dependent interventions (e.g., a specific suggestion or strategy) could confer behavioural control over these illusions. Future studies could test whether explicitly providing potentially effective strategies, such as those shown to reduce Stroop interference (Palfi et al., 2022), reduces susceptibility to these illusions, and whether the present instructions are sufficient to reduce interference in tasks in which cognitive penetration is known to be possible.**"

Minor comment

Please change descriptions of procedures and analyses from future to past tense.

We thank Dr Palfi for this comment. All descriptions of completed procedures and analyses are now in the past tense (see also our response to the Recommender, Comment 9). The only exception is the study design table (Table 1), which is reproduced as approved at Stage 1.

Review by Reshanne Reeder

2A. Whether the data are able to test the authors' proposed hypotheses (or answer the proposed research question) by passing the approved outcome-neutral criteria, such as absence of floor and ceiling effects or success of positive controls or other quality checks.

There are a lot of inconclusive Bayes Factors [edit: though it is very hard to see in the figure, and the results are not actually reported in the text anywhere], putting into questions whether the sample size was large enough. The ~60-participant difference between what was proposed in the methods vs. what was collected in actuality may have contributed somewhat to this lack of evidence in either direction.

We thank Dr Reeder for this comment. As the Recommender notes in their assessment of criterion 2A, the registered sampling plan specified the recruitment of around 500 participants, which we achieved. We acknowledge, however, that applying all registered exclusion criteria (in particular the space bar attention check; see our response to the Recommender, Comment 4) reduced the sample of the fully registered analysis, which is now the main analysis, to 295 participants. This reduction is larger than in the initial Stage 2 submission, and we now discuss its implications for the strength of the evidence as a limitation.

All results are now reported in the text and in Table 3. In the fully registered analysis, 10 of the 14 registered tests provided evidence for the null, 4 were inconclusive, and none supported a relationship. The inconclusive results are concentrated in the Vertical–Horizontal illusion and in the secondary outcome (IES). In the unregistered sensitivity analysis, with a larger sample (N = 458 per illusion), 12 of 16 tests provided evidence for the null, 4 were inconclusive, and none supported a relationship, and the conclusions for the primary outcome were unchanged (Table A1 and A2).

Lines 540-546:
"**The sensitivity of the tests was further limited by the sample size of the registered analysis: applying the registered space bar check reduced the sample from 468 to 295 participants, and the inconclusive results (concentrated in the Vertical-Horizontal illusion and the secondary outcome) may partly reflect this reduced sensitivity rather than evidence for a relationship. Notably, the unregistered analysis, with a larger sample, led to the same conclusions for the primary outcome.**"

2B. Whether the introduction, rationale and stated hypotheses (where applicable) are the same as the approved Stage 1 submission. This can be readily assessed by referring to the tracked-changes manuscript supplied by the authors.

I can confirm the introduction has not changed substantially between IPA and Stage 2. However, "positive association" has been replaced with "negative association" in the 'interpretation given different outcomes' section of Table 1 (hypothesized direction of effects), with no explanation provided for this change.

We thank Dr Reeder for this careful check. The direction of this hypothesis was corrected during Stage 1 review, in correspondence with the Recommender, immediately before in-principle acceptance, and the Recommender has confirmed this in the decision letter. The "negative association" in Table 1 therefore reflects the approved Stage 1 hypothesis. We have also corrected the remaining references to a positive association in the Results and Discussion, so that the hypothesised

direction is now described consistently as negative throughout the manuscript (see our response to the Recommender, Comment 9).

2C. Whether the authors adhered precisely to the registered study procedures.

Methods:

Participants:

The final sample size aim of "around 500" was not reached (N=444-446).

We thank Dr Reeder for this comment. The target of "around 500" referred to recruitment, and 500 participants were recruited, as planned; as the Recommender notes in their assessment of criterion 2A, the Stage 1 protocol did not require a final sample of 500 after exclusions. The final samples are now 295 participants in the fully registered (main) analysis, with the same participants included for all three illusions, and 468 participants in the unregistered sensitivity analysis (458 analysed for each illusion). All exclusions are reported in the Results and in a flow diagram (Figure A1; see our response to the Recommender, Comment 1). The implications of the reduced sample of the registered analysis are discussed in our response to criterion 2A above.

Line 140: "English native speakers" has changed to "English speaking adults" – how many of these were native speakers? Was level of English proficiency assessed in some way?

We thank Dr Reeder for spotting this. Not all participants were native English speakers: based on participants' Prolific demographic data, 59 of the 500 recruited participants (11.8%), and 43 of the 295 participants in the fully registered analysis (14.6%), did not report English as their first language. Eligibility was implemented using Prolific's screening filters, which admitted participants whose first or primary language was English; English proficiency was not otherwise assessed.

The Stage 1 description ("English native speakers") did not reflect this criterion and was an error in the Stage 1 manuscript. This is now reported as a deviation from the Stage 1 protocol. Restricting the registered analysis to native English speakers (N = 252), as specified at Stage 1, did not change the conclusions for the primary outcome (Table A1 and A2). Please see our response to the Recommender, Comment 9, for details.

Line 144: The words "consent form" have been deleted – was there no consent form in the actual study?

We thank Dr Reeder for noting this. A consent form was used, as approved: participants were presented with an information sheet and consent form at the start of the experiment, and could only proceed by clicking "I read, understood, and I consent". The deletion was unintentional, and the Stage 1 wording has been restored, changed only to the past tense (see our response to the Recommender, Comment 9).

Procedure:

Line 162: The words “(thus limiting potential over-generalization)” have been added.

We thank Dr Reeder for noting this. The phrase has been removed, so that the Methods match the approved Stage 1 text.

The time it took for participants to complete the Illusion Game in the original study has been changed from 55 to 48 minutes (this should not change, as it is referring to an earlier study?).

Dr Reeder is correct, and we thank them for spotting this. The original Illusion Game lasted approximately 55 minutes; the 48 minutes was the median completion time of the present study (47.96 minutes), which was inserted into this sentence by mistake. The Stage 1 figure has been restored (see our response to the Recommender, Comment 9).

Registered protocol attention check exclusion criterion change: the choice to remove the exclusion criterion that participants should be removed if they do not press the spacebar 5 times during the attention check, is acceptable. I believe that technical lags during online data collection can certainly lead to dropped multiple responses in a row.

We thank Dr Reeder for this comment. On closer inspection, 177 of the 189 participants who failed this check did not press the space bar at all, which argues against responses lost due to technical lags (which would produce partial counts). Most of them (181 of 189) passed the other two attention checks, and their scores on the remaining PCS items were not lower than those of participants who passed ($M = 2.49$ vs. 2.28; $BF_{10} = 2.90$). We believe instead that the failure rate reflects how the check was implemented: the “PLEASE WAIT” screen appeared as soon as the recording ended, while participants were being told to open their eyes, and only recorded presses made while it was displayed (see our response to the Recommender, Comment 4).

Registered protocol Illusion game changes: removing participants with extreme error rates in the congruent condition seems sensible, though should justify the specific 97.5% CI boundary. But why not use the registered exclusion criterion (>45% error rate)? Further down in Table 2, mean error rates exceed the 45% cutoff in the “strong” condition, but the methods state: “any participant exhibiting an error rate greater than 45% for any illusion type will be considered to be responding at chance level (i.e., randomly) and flagged accordingly”, with block-level exclusions further specified. Why was this not done here?

We thank Dr Reeder for this comment. The registered criterion (> 45% error rate) was applied, but the initial Stage 2 submission did not report it separately. In the fully registered (main) analysis, 4 participants exceeded it in the first block and were fully excluded, and 3 exceeded it only in the second block and had that block removed (8 and 7, respectively, in the sample used for the unregistered sensitivity analysis). These exclusions are now reported in the Results and in a flow diagram (Figure A1; see our response to the Recommender, Comment 1).

The high group means in the Strong condition (Table 2) reflect the intended effect of strong illusions on average performance. The registered criterion is computed per participant and per block across all trials of an illusion type, not per condition, so these means do not indicate that individual participants were responding at chance level.

The registered criterion and the additional exclusion address different problems. The registered criterion identifies participants responding at chance level across all trials of an illusion type. The additional exclusion targets participants whose errors were concentrated in the congruent condition, where the illusion makes the task easier and errors should be rare, a pattern more consistent with a misunderstanding of the instructions than with illusion sensitivity, and which the registered criterion does not necessarily detect. The 97.5% level was chosen to flag only the most extreme responses; as it was not specified in advance, it is applied only in the unregistered sensitivity analysis, and removing it did not change the results (Tables A1 and A2; see our response to the recommender, Comment 2).

I am actually a little confused as to where deviations from protocol end and the registered analysis begins. Please designate this in the manuscript more clearly.

We agree that this was unclear. The fully registered analysis, which applies only the procedures of the approved protocol, is now the main analysis, and the Results are organised so that registered procedures, deviations and exploratory analyses are clearly separated. Registered Exclusions reports the exclusions of the approved protocol (N = 295). Deviations From Registration describes each deviation and its rationale.

(1) Correlations → Registered Analysis reports the results under the approved protocol (Table 3, Figure 2).

(2) Correlations → Sensitivity Analyses evaluates the consequences of the deviations, both combined and one at a time (Table A1 and A2).

(3) Exploratory Analyses reports the analyses that were not registered.

The flow diagram (Figure A1) also distinguishes registered exclusions from deviations (see our response to the Recommender's general note on the organisation of the Results).

The final sample sizes listed on the GitHub page are ~20 participants more than what is reported in the manuscript.

We thank Dr Reeder for spotting this. The discrepancy arose because the online report gave the sample size before the per-illusion exclusion of participants with extreme error rates, whereas the manuscript gave it after. This exclusion is no longer applied in the main analysis. The manuscript and the online reports now report the same samples: 295 participants in the fully registered (main) analysis, and 468 participants (458 per illusion after the per-illusion exclusion) in the unregistered sensitivity analysis (see our response to the Recommender, Comment 1).

2D. Where applicable, whether any unregistered exploratory analyses are justified, methodologically sound, and informative.

Analyses were repeated with the data collapsed across illusion strength and also split by difficulty level. These are fine exploratory analyses. Perhaps due to the number of inconclusive BFs, a bootstrapping procedure could supplement exploratory analyses.

We thank Dr Reeder for this suggestion. We note that, following the Recommender's Comment 5, collapsing across illusion strength now follows the registered decision rule and is part of the main analysis; the exploratory analyses are now the analysis with Mild and Strong kept separate and the split by difficulty level (Exploratory Analyses).

As the analyses are Bayesian, we report the posterior median and 95% credible interval of every correlation alongside its Bayes factor (Table 3), which quantify the uncertainty in each estimate that a bootstrapping procedure would otherwise be used to estimate. In addition, the sensitivity analyses (Table A1 and A2) show how the classification of each set of tests changes across preprocessing choices and sample sizes. We therefore did not add a bootstrapping procedure, but we believe these additions address the stability of the inconclusive results.

2E. Whether the authors' conclusions are justified given the evidence.

Results:
Line 366: This line still states that the hypothesized relationship is positive. Is the hypothesized relationship meant to be negative now?

We thank Dr Reeder for spotting this; it has been corrected. The registered hypothesis is a negative association, and the Methods, Results and Discussion now state this consistently (see our response to the Recommender, Comment 9).

Paragraph lines 357-363: Any inconclusive BFs, and BFs generally "below 3" are not evidence for the hypothesized null effect. The hypothesized null effect would only be supported by Bayesian evidence supporting a true null effect (BF10 < .33).

We agree. Tests are now counted as supporting the null only when $BF_{10} \leq 1/3$, and inconclusive results are reported as such (see Recommender Comments 7 and 8).

Descriptives for PCS and psychoticism are missing. All BFs for all analyses are missing.

We thank Dr Reeder for this comment. The mean, SD and range of PCS and psychoticism scores, and the skewness of psychoticism, are now reported in the Results, and their distributions are shown in Figure A2.

All Bayes factors for the registered analyses are now reported in Table 3, together with the correlation estimates and their 95% credible intervals. For the sensitivity analyses, Table A1 reports the number of tests in each evidential category and Table A2 the Bayes factor of every test; the individual Bayes factors of the exploratory analyses are reported in the analysis report.

Discussion:

In the first couple paragraphs of the Discussion, it is stated that Bayesian analyses suggested an absence of a relationship, whereas some of the results were inconclusive in either direction. It would appear there is an entire table or paragraph missing from the results section (where are all the BFs?), so it is very hard to tell which tests actually showed evidence for an absence and which were inconclusive.

We agree, and thank Dr Reeder for this comment. All Bayes factors for the registered tests are now reported in the Results text and in Table 3 (see our response to the Recommender, Comment 7), and

the Discussion now distinguishes the tests that provided evidence for the absence of a relationship from those that were inconclusive (see our response to the Recommender, Comment 8).

For the primary outcome, the evidence favoured the absence of a relationship between PCS and error rates for the Müller-Lyer and Ebbinghaus illusions but was inconclusive for the Vertical–Horizontal illusion, and favoured the absence of a relationship between psychoticism and error rates for all three illusions; for the secondary outcome (IES), the evidence was mixed.

Paragraph starting line 400: Did you also check the distribution of psychoticism scores? Maybe there just wasn't much variability in your sample. It would be nice to include (either in supplementary material or in the main text) descriptives on the distributions for psychoticism (and also PCS), and comment on these in the Discussion to support discussion points.

We thank Dr Reeder for this suggestion. PCS scores (possible range 0–5) had a mean of 2.16 (SD = 0.80, range: 0.00–4.20), and psychoticism scores (possible range 0–3) a mean of 0.48 (SD = 0.40, range: 0.00–1.80), with a positively skewed distribution (skewness = 0.85). These descriptives are now reported in the Results, and both distributions are shown in Figure A2. The Discussion now considers the restricted variability of psychoticism as one possible contributor to the absence of an association. We also note that, for the primary outcome, all psychoticism Bayes factors fell below 1/3, i.e., the data favour the absence of an association rather than being merely uninformative.

Lines 494-503:
"In contrast to our registered hypothesis, we found no evidence of an association between visual illusion sensitivity and the psychoticism facet of the PID-5**; for the primary outcome, the Bayes factors favoured the absence of an association rather than being merely uninformative. […] One possible contributor is the limited variability of psychoticism in our general-population sample: scores were low on average (M = 0.48 on a 0–3 scale) and positively skewed, with most participants scoring near the bottom of the scale (Figure A2). Such a restricted range limits the ability to detect associations.**"

Misc. comments

Figure 2: This figure is somewhat unclear. First of all, dashed lines overlap tiny font for BFs, obscuring the value to the left of the decimal point in most cases. It is unclear what the X axis is, as it has no label. Why is there a gap in the reddish shading halfway down the table? Perhaps IES and error rates should be separated into 2 separate tables. What does the BF in gray represent (not in the key)?

We thank Dr Reeder for these comments, which have helped us improve the figure substantially. Figure 2 has been revised as follows. The BF labels, previously placed on the zero line where the dashed reference line obscured them, are now placed at the right of each panel, clear of the reference line, and are reported to two decimal places. The x-axis is now labelled ("Correlation (posterior median and 95% credible interval)"). The shading has been removed; the "gap" in the shading was the spacing between panels. Error rate and IES are now shown in separate rows of panels, labelled as the primary and secondary outcomes, with more space between them. The grey BF labels corresponded to the scores collapsed across Mild and Strong, which had not been assigned a colour; these scores now have their own colour and legend entry ("Mild + Strong"). The caption explains all labels and symbols, and all values for the registered tests are also reported in Table 3. Finally, as noted by the Recommender (Comment 6), the figure now displays the intended analysis (the fully registered analysis).